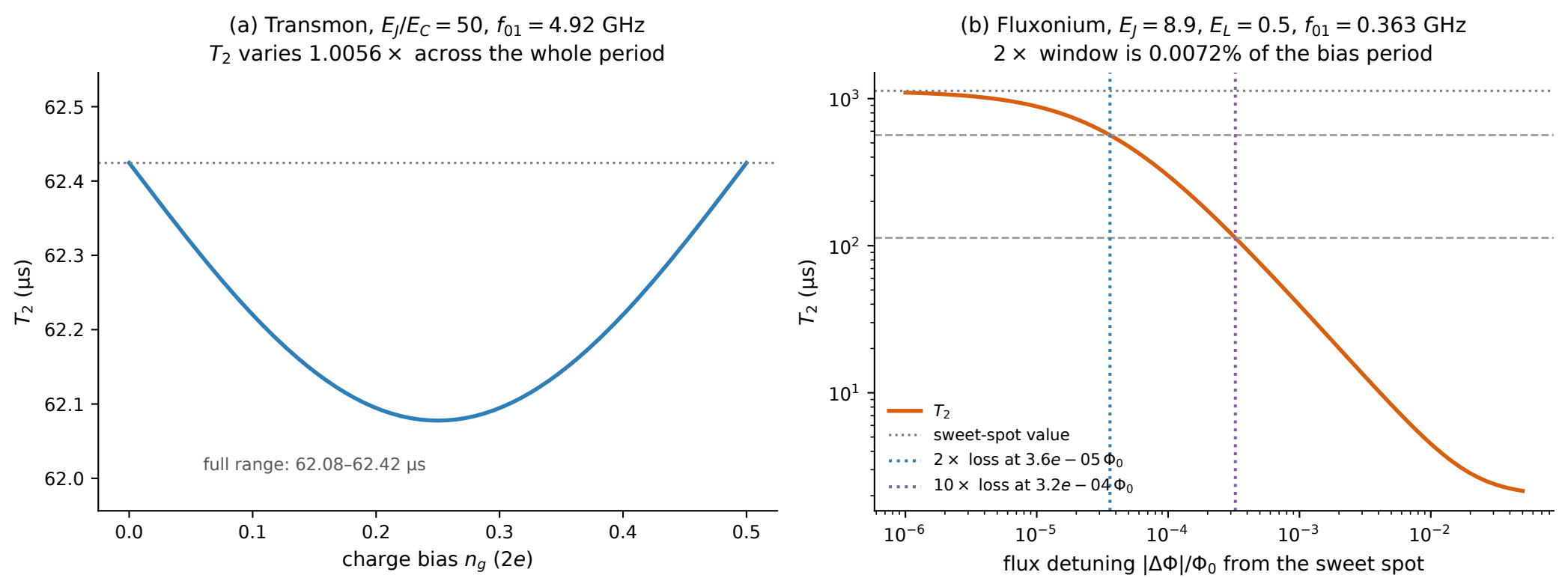

(a) Transmon, $E_J/E_C = 50$, $f_{01} = 4.92$ GHz
T_2 varies 1.0056 × across the whole period
62.5
62.4
62.3
62.2
62.1
62.0
T_2 (μs)
full range: 62.08–62.42 μs
0.0
0.1
0.2
0.3
0.4
0.5
charge bias n_g ($2e$)
(b) Fluxonium, $E_J = 8.9$, $E_L = 0.5$, $f_{01} = 0.363$ GHz
2 × window is 0.0072% of the bias period
10^3
10^2
10^1
T_2 (μs)
T_2
sweet-spot value
2 × loss at $3.6e-05\,\Phi_0$
10 × loss at $3.2e-04\,\Phi_0$
10^{-6}
10^{-5}
10^{-4}
10^{-3}
10^{-2}
flux detuning $|\Delta\Phi|/\Phi_0$ from the sweet spot